

Exact maps and analytic certificates

Key advances in the continuing reconstruction

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The starting point is the claimed complex-threefold construction on S^6 circulated by Levent Alpöge with Fable. The continuing audit has produced explicit results about its analytic fibration and about the marked Niemeier–Leech–triality constructions investigated alongside it. This companion gives the actual formulas and proof locations for a selection of those advances. It does not replace the complete arguments.

1. The anticanonical ring recovers the fibration

Let $f : X \rightarrow B = \mathbb{P}^1$ denote the analytic construction with its original periods and fillings. Its finite reduced fibres S_1, S_2 and base coordinate t satisfy

$$f^*(p_1) = 3S_1, \quad f^*(p_2) = 4S_2, \quad t(p_1) = 0, \quad t(p_2) = 1, \quad t(p_0) = \infty.$$

The evaluation $f^*f_*\omega_{X/B} \rightarrow \omega_{X/B}$ has zero divisor *exactly* $2S_2$. The resulting identities are

$$\begin{aligned} \omega_X &\simeq \mathcal{O}_X(-2S_2), & \omega_X^{-2} &\simeq f^*\mathcal{O}_B(p_2), \\ f_*\mathcal{O}_X(kS_2) &= \mathcal{O}_B(\lfloor k/4 \rfloor p_2) & (k \in \mathbb{Z}). \end{aligned}$$

The last equality is equality of subsheaves of meromorphic functions: on the original cover $t_2 = s_2^4$, an invariant Laurent series contains only powers divisible by four. The pole condition is precisely $4 \operatorname{ord}_{p_2}(b) + k \geq 0$.

Consequently the *full graded algebra*, not merely its dimensions, is

$$\bigoplus_{m \geq 0} H^0(X, \omega_X^{-m}) \simeq \mathbb{C}[U, V], \quad \deg U = 1, \quad \deg V = 2.$$

Here $U \mapsto u$ with $\operatorname{div}(u) = 2S_2$, and $V \mapsto v = f^*\ell$ under the specified degree-two identification, where $\ell \in H^0(B, \mathcal{O}_B(p_2))$ is nonzero at p_2 . In degree m , the basis is $u^{m-2j}v^j$ for $0 \leq j \leq \lfloor m/2 \rfloor$. Writing $\ell = A/(t-1) + B'$ with $A \neq 0$, the complete degree-two map is

$$x \longmapsto [t(f(x)) - 1 : A + B'(t(f(x)) - 1)].$$

Thus the fibration is recovered intrinsically from X , up to this explicitly retained projective-coordinate choice. This is what makes it possible to test a deformation of the whole threefold, rather than just a marked torus fibre.

Full proof: *Complete current workbench*, Theorem 53.17, pp. 699–700; source `analytic_canonical_ring.tex`, equations CR1–CR5.

2. The original period constant gives a nonzero deformation

Keep the cusp-regular period solution β^{part} and put

$$\beta_c = \beta^{\text{part}} + c, \quad D_{\text{part}} = \text{Im } \beta^{\text{part}} - \frac{6(\text{Im } \mu)^2}{\text{Im } \tau}, \quad M = \max_{B \setminus \{p_0\}} D_{\text{part}}.$$

On every open disc Q whose closure lies in $\{c \in \mathbb{C} : \text{Im } c < -M\}$, the original charts and gluing maps give a proper holomorphic submersion $\mathcal{X}_Q \rightarrow Q$ with fibres X_c . The construction retains all three critical values $0, 1, \infty$.

At a fixed lifted regular base point, fix $c_* \in Q$ and write

$$T = \text{Im } \tau > 0, \quad m = \text{Im } \mu, \quad D = \text{Im } \beta_{c_*} - 6m^2/T < 0, \quad P = \begin{pmatrix} 0 & 0 \\ -m/T & 1 \end{pmatrix}, \quad \delta = c - c_*.$$

The unchanged ordered period matrix is

$$\Pi_c = \begin{pmatrix} 6\mu & \tau & 1 & 0 \\ \beta_c & \mu & 0 & 1 \end{pmatrix}, \quad \Pi_c : \mathbb{R}^4 \longrightarrow \mathbb{C}^2.$$

Its inverse below is real-linear. The exact marked map between the two period lattices is

$$w = \left(I + \frac{\delta}{2iD} P \right) \zeta - \frac{\delta}{2iD} P \bar{\zeta}, \quad \det_{\mathbb{R}}(\Pi_c \Pi_{c_*}^{-1}) = \frac{D + \text{Im } \delta}{D} > 0.$$

In the convention that the transported $(0, 1)$ fields are $\partial_{\bar{\zeta}_k} + \sum_j \nu_k^j \partial_{\zeta_j}$,

$$\nu_\delta = \frac{\delta}{2iD + \delta} P.$$

With the connecting-class convention $\text{KS}(\partial_c) = [\bar{\partial}h]$, the regular-fibre Kodaira–Spencer class is

$$- \left[\frac{1}{2iD} \partial_{\zeta_2} \otimes \left(d\bar{\zeta}_2 - \frac{m}{T} d\bar{\zeta}_1 \right) \right] \neq 0.$$

Its constant nonzero coefficient cannot be a derivative of a periodic coefficient: integrating such a derivative over the torus gives zero.

The additional *global* step is essential. Over $A = \mathbb{C}[\epsilon]/(\epsilon^2)$, the intrinsic degree-two anticanonical sections form a free A -module of rank two. A first-order trivialization of X_c reducing to the identity at c_* would identify the fibrations through an element of $\text{PGL}_2(A)$. The actual critical schemes fix the three constant sections $0, 1, \infty$, forcing that element to be the identity. Restriction to a regular fibre then contradicts the displayed nonzero class. Hence

$$\text{KS}_{X_{c_*}}(\partial_c) \neq 0 \quad \text{in } H^1(X_{c_*}, T_{X_{c_*}}).$$

This does not assert that every two parameters give nonisomorphic unmarked manifolds, or that the family exhausts all deformations.

Full proof: Workbench, Theorem 53.18, pp. 701–705; `period_parameter_deformation.tex`, PD1–PD18.

3. Finite fillings: explicit covers and the full attachment kernel

At the two original finite centres retain

$$\rho = e^{\pi i/3}, \quad \mu_1 = (2 - \rho)/3, \quad \mu_2 = (1 - i)/2, \quad \kappa_1 = \beta_1 + 2\mu_1, \quad \kappa_2 = \beta_2 + 3\mu_2,$$

with $\text{Im } \kappa_j < 0$. Set

$$E_j = \mathbb{C}/(\mathbb{Z} + \kappa_j \mathbb{Z}), \\ F_1 = \mathbb{C}/(\mathbb{Z}(\rho - 2) + \mathbb{Z}(-3)), \quad F_2 = \mathbb{C}/(\mathbb{Z}(i - 1) + \mathbb{Z}(-2)).$$

The maps from $E_j \times F_j$ to the original central torus are induced by

$$L_1(\xi, \eta) = (\eta, \xi - \eta/3), \quad L_2(\xi, \eta) = (\eta, \xi - \eta/2).$$

Their kernels are generated respectively by $(2/3, \rho)$ and $(1/2, i)$, of orders three and two. The lifted affine generators are

$$g_1(\xi, \eta) = (\xi + \kappa_1/3, (\rho - 1)\eta), \quad g_2(\xi, \eta) = (\xi - \kappa_2/4, -i\eta).$$

Together with the kernel translations they act freely, giving product covers of the central surfaces of degrees nine and eight. The varying cover is identified with the normalization of the *actual* ramified base change, including every branch of $s_j^{m_j} = u^{m_j}$.

Let Λ now denote the original rank-four fibre lattice, and let A_j, v_j, m_j be its finite linear monodromy, translation vector and order, with $(m_1, m_2) = (3, 4)$ and $A_j v_j = v_j$. The complete deck-group presentation and attachment kernel are

$$\Gamma_j = \langle \Lambda, G_j \mid G_j t_\lambda G_j^{-1} = t_{A_j \lambda}, \quad G_j^{m_j} = t_{v_j} \rangle, \\ \ker(\Lambda \rtimes_{A_j} \mathbb{Z} \longrightarrow \Gamma_j) = \langle \tilde{G}_j^{m_j} t_{-v_j} \rangle \simeq \mathbb{Z},$$

and this kernel is central. The computation determines the entire kernel, not merely a relation that lies in it.

Finally let $\gamma \in \Lambda^*$ be the retained invariant coordinate, $\varepsilon_j = \gamma(v_j) \in \{1, -1\}$ and $\zeta_j = e^{-2\pi i/m_j}$. The normal line N_{S_j/N_j} has exact holomorphic order m_j , but the explicit nonzero smooth section is

$$H_j(x) = \exp(-2\pi i \varepsilon_j \gamma(x)), \quad H_j(A_j x + v_j/m_j) = \zeta_j H_j(x).$$

The smooth trivialization is $[x, n] \mapsto ([x], n/H_j(x))$, with inverse $([x], z) \mapsto [x, zH_j(x)]$.

Full proof: Workbench, Theorems 53.4, 53.6, 53.7 and 53.9, pp. 684–690; `finite_filling_certificates.tex`, FF1–FF46.

4. Niemeier–Leech transport retains the cubic arithmetic

Start with the marked root lattice $R = A_5^{\perp 4} \perp D_4$ and its Niemeier overlattice N , with $[N : R] = 72$. Its rank is 24; 72 is the glue index, not a dimension. For its original Weyl vector ρ_N and retained root α , put

$$K = \{x \in N : (\rho_N, x) \equiv 0 \pmod{6}\}, \quad \Lambda_L = K + \mathbb{Z}(\rho_N/6 - \alpha).$$

Then $N \cap \Lambda_L = K$ and $[N : K] = [\Lambda_L : K] = 6$. The residue order-three action cycles three such neighbours. Their common pairwise intersection J has determinant 81 and index 9 in each neighbour. The fourth extension M , fixed by this action, has root system A_2^{12} and 72 roots. These are explicit integral constructions, retaining the original residue marking and glue generators.

Use the fixed real isometry $\mathcal{L}$ to $\mathbb{O}^3$ with the original Cayley frame $(1, i, j, k, \ell, j\ell, k\ell)$. Put

$$\tau(q) = \text{Re}((q_1 q_2) q_3), \quad F = \tau \circ \mathcal{L}, \quad \mathcal{V}_F(\mathcal{A}) = \langle F(x) : x \in \mathcal{A} \rangle_{\mathbb{Z}}.$$

For a lattice $\mathcal{A}$, this defines an *additive group generated by values*; it does not identify the nonlinear image $F(\mathcal{A})$ with that group. For the retained cubic, the complete calculation gives

$$\begin{aligned} \mathcal{V}_F(N) &= \frac{1}{27}\mathbb{Z} \oplus \frac{\sqrt{2}}{54}\mathbb{Z} \oplus \frac{\sqrt{3}}{18}\mathbb{Z}, \\ \mathcal{V}_F(J) &= \mathcal{V}_F(M) = \frac{1}{27}\mathbb{Z} \oplus \frac{\sqrt{2}}{216}\mathbb{Z} \oplus \frac{\sqrt{3}}{18}\mathbb{Z}, \\ \mathcal{V}_F(\Lambda_L) &= \frac{1}{27}\mathbb{Z} \oplus \frac{\sqrt{2}}{216}\mathbb{Z} \oplus \frac{\sqrt{3}}{54}\mathbb{Z}. \end{aligned}$$

In the displayed ordered generators the two group inclusions have matrices

$$\mathcal{V}_F(N) \hookrightarrow \mathcal{V}_F(J) : \text{diag}(1, 4, 1), \quad \mathcal{V}_F(J) \hookrightarrow \mathcal{V}_F(\Lambda_L) : \text{diag}(1, 1, 3).$$

Their quotients are $\mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$; the total quotient is $\mathbb{Z}/12\mathbb{Z}$. These are inclusions of *value groups*, not a claim that $N \subset \Lambda_L$. The proof gives both integral containment and reverse generator certificates using complete cubic polarization.

For the same embedding into the Albert algebra, the equal-diagonal determinant is

$$\det Q_{\mathbb{O}}(\lambda; \mathcal{L}y) = \lambda^3 - \lambda(y, y) + 2F(y).$$

The coefficient 2 remains part of the map. Beyond the lattice, with $E(q) = Q_{\mathbb{O}}(0; q)$, the spectral graph

$$q \mapsto Q_{\mathbb{O}}(-\lambda_{\min}(E(q)); q)$$

maps onto the equal-diagonal positive-semidefinite determinant boundary, with inverse given by its ordered off-diagonal coordinates. The associated Peirce products retain their coefficient $1/2$.

Full proof: Higher torus wrapping and the octonionic 24-dimensional continuation, Sections 9–11, pp. 106–125; spectral graph Theorem 5.12, p. 34. Source labels: `nca:sec:cubic-arithmetic`, `esrf:sec:fixed-neighbor`, `jm:sec:JM-extension`.

5. A literal common lattice in the original triality space

Wilson’s octonionic Leech construction is an established input, not a new construction claimed here. Retain its norm $q_W(x) = \frac{1}{2} \sum_{j=1}^3 |x_j|^2$ and the isometry $\iota(x) = x/\sqrt{2}$ into the Euclidean norm $\sum |q_j|^2$. Write $\Lambda_{\text{cyc}} = \iota(\Lambda_W)$. The new comparison computes both literal intersections:

$$\Lambda_{\text{cyc}} \cap \mathcal{L}(N) = \Lambda_{\text{cyc}} \cap \mathcal{L}(\Lambda_L) = \mathcal{I} = \{\sqrt{2}(a, a, a) : a \in D_4\},$$

where here $D_4 = \{a \in \mathbb{Z}^4 : \sum_{i=0}^3 a_i \equiv 0 \pmod{2}\}$ lies in the *first four original Cayley coordinates*. The Gram matrix, determinant and minimum squared norm of $\mathcal{I}$ are

$$6G_{D_4}, \quad 5184, \quad 12.$$

The cubic on this common lattice is explicit:

$$\tau(\sqrt{2}(a, a, a)) = 2\sqrt{2} \underbrace{a_0(a_0^2 - 3(a_1^2 + a_2^2 + a_3^2))}_{P(a)}.$$

The generated group is $4\sqrt{2}\mathbb{Z}$, but $12\sqrt{2}$ is not attained. Indeed $P(a) \equiv a_0^3 \pmod{3}$, so $3 \mid P(a)$ forces $3 \mid a_0$; substitution then gives $9 \mid P(a)$. The required value $P(a) = 6$ is impossible. This gives a concrete arithmetic distinction between a group of values and individual values, while retaining their exact common embedding.

Full proof: Higher-rung paper, Section 12, pp. 125–135; `es_niemeier_wilson_cyclic.tex`. Primary input: R. A. Wilson, *Octonions and the Leech lattice*, J. Algebra **322** (2009), 2186–2190, [published paper](#); [author version](#), Sections 2–4.

Reading the release

The *complete higher-rung paper* contains the triality and lattice proofs. The *complete current workbench* also contains the analytic ring, deformation and finite-filling arguments. Printed page numbers above refer to the 6 September copies (138 and 782 pages respectively).

The separately labelled *complete-project calculations and provenance ZIP* retains the full TeX, exact checkers and results, mathematical datasets, standalone papers and mathematical history, including work not selected for this front. The older published files remain unchanged as dated records. All proof-source filenames above lie under `project/supporting_materials/workbench/research/` in that archive.

These advances do not by themselves certify every global step in the S^6 claim or establish a counterexample to CDP20. No interacting quantum-field-theory mass-gap conclusion follows from the displayed lattice maps. The complete workbench retains the arguments and their dependencies for further checking; this is an AI-assisted research record, not independent expert adjudication or Lean formalization.

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